NATURAL LANGUAGE PROCESSING
ASSIGNMENT – 04

Details	Information
Course	Natural Language Processing
Course Outcome	CO 4 – Semantic representation, interpretation and meaning-based NLP systems
Assignment	Assignment 04 – Part I and Part II
Total	100 marks (50 marks each part)
Student Name	V.Lakshmi Prasad
Register Number	192425179
Submission Date	

PART I – SEMANTIC ANALYSIS THROUGH STRUCTURED REASONING

Part I follows the requested progression: interpret the text, identify semantic units, choose a representation, analyse meaning, resolve ambiguity and evaluate.

Task 1 – Meaning Representation and Language Structure (8 Marks)

1.1 Identify Meaning Components

Component	Answer
Entities	Maya (person), Ravi (person), a book (object), yesterday (time expression).
Event	GIVE – a transfer event.
Agent / giver	Maya – the person who performs the giving.
Recipient	Ravi – the person who receives the book.
Theme	A book – the object transferred.
Time	Yesterday – the event occurred on the day before the utterance day.

Linguistic form: a declarative English sentence with subject–verb–indirect object–direct object–time adjunct order. Meaning: Maya performed a giving event, the book was the transferred object, Ravi was the recipient, and the event happened yesterday.

1.2 Compare Meaning Representations

A. Conceptual / event structure:

GIVE(event_1) | Agent(event_1, Maya) | Theme(event_1, Book_1) | Recipient(event_1, Ravi) | Time(event_1, Yesterday)

B. Frame / slot representation:

Frame: GivingEvent	Value
Event ID	event_1
Action	GIVE
Agent	Maya
Theme	a book
Recipient	Ravi
Time	yesterday

The event structure makes the action and semantic roles explicit in predicate form. The frame stores the same information in named slots and is convenient for software systems. Neither representation identifies the exact book, the exact calendar date, or whether the book was physically handed over; those details are not stated.

1.3 Meaning Structure and Ambiguity

Word order helps identify grammatical relations, but semantic roles describe the meaning independently of position. In the original sentence, Maya is the agent, Ravi is the recipient and the book is the theme.

Meaning-preserving variation: “Yesterday, a book was given to Ravi by Maya.” The surface order and voice change, but the semantic representation remains: GIVE(event_1), Agent(event_1, Maya), Theme(event_1, Book_1), Recipient(event_1, Ravi), Time(event_1, Yesterday).

1.4 Design Defence

This representation is suitable for a computer because it converts a sentence into labelled entities, an event and explicit semantic-role relations. A program can store, query and compare these fields without relying on the original word order. It can answer questions such as who gave the book, what was given, to whom and when. Missing values can also be represented as unknown instead of being guessed.

Task 2 – First Order Predicate Calculus (8 Marks)

2.1 Define the Logical Vocabulary

Symbol	Meaning / argument order
Student(x)	x is a student.
Course(y)	y is a course.
Instructor(z)	z is an instructor.
Enrolled(x,y)	Student x is enrolled in course y.
Teaches(z,y)	Instructor z teaches course y.
Passed(x,y)	Student x passed course y.
Constants	For example, rao denotes Dr. Rao; nlp denotes the NLP course.
Variables	x, y and z range over students, courses and instructors as appropriate.

2.2 Translate Natural Language into Logic

(i) Every student is enrolled in at least one course:

$\forall x\,[\text{Student}(x) \rightarrow \exists y\,(\text{Course}(y) \wedge \text{Enrolled}(x,y))]$

(ii) Some student passed every course taught by Dr. Rao:

$\exists x\,[\text{Student}(x) \wedge \forall y\,((\text{Course}(y) \wedge \text{Teaches}(rao,y)) \rightarrow \text{Passed}(x,y))]$

(iii) No instructor teaches every course:

$\forall z\,[\text{Instructor}(z) \rightarrow \neg\forall y\,(\text{Course}(y) \rightarrow \text{Teaches}(z,y))]$

Equivalent form for (iii): $\forall z\,[\text{Instructor}(z) \rightarrow \exists y\,(\text{Course}(y) \wedge \neg\text{Teaches}(z,y))]$.

2.3 Interpret and Check the Formulae

For formula (i), the outer universal quantifier ∀x means the statement applies to each student. For each such student, the inner existential quantifier ∃y requires at least one course in which that student is enrolled. The course may be different for different students.

Quantifier order can change meaning. Compare ∀x ∃y Enrolled(x,y) ("each student has some course") with ∃y ∀x Enrolled(x,y) ("there is one course in which every student is enrolled").

Finite model	Facts	Result
Students: A, B; Courses: C1, C2	Enrolled(A,C1), Enrolled(B,C2)	∀x∃y is true; ∃y∀x is false.

Therefore, the two formulae are not logically equivalent. The first permits different courses for different students; the second requires a shared course.

2.4 Logical Consistency

The pair cannot both be true if "NLP" denotes the same course and the domain of students is the same. "Every student passed NLP" means ∀x(Student(x) → Passed(x,nlp)). "There is a student who did not pass NLP" means ∃x(Student(x) ∧ ¬Passed(x,nlp)). The second formula directly contradicts the first for that student.

Task 3 – Linguistically Relevant Concepts and Syntax-Driven Semantic Analysis (9 Marks)

3.1 Identify Linguistically Relevant Concepts

Concept	Role in the sentence
The experienced researcher	Entity / subject; head noun researcher, modified by experienced.
saw	Predicate / perception event.
the student	Entity; possible object/theme of seeing.
with the telescope	Prepositional phrase; can describe the instrument used by the researcher or the student's accompaniment/possession.
Semantic roles	Experiencer/agent = researcher; perceived entity = student; instrument = telescope in the instrument reading.

These concepts matter because the system must identify who saw whom and how the telescope is related to the event or participant.

3.2 Syntax-to-Semantics Mapping

Attachment A – instrument attachment to the verb phrase: [researcher [saw [the student] [with the telescope]]]. Meaning: the researcher used a telescope to see the student.

Representation: SEE(e1) ∧ Experiencer(e1,researcher) ∧ Theme(e1,student) ∧ Instrument(e1,telescope).

Attachment B – noun attachment to “student”: [researcher saw [the student [with the telescope]]]. Meaning: the researcher saw a student who had or was carrying the telescope.

Representation: SEE(e1) ∧ Experiencer(e1,researcher) ∧ Theme(e1,student) ∧ Has(student,telescope).

A syntax-driven parser first builds phrase structure, identifies the prepositional phrase and its possible attachment node, then composes the corresponding semantic relation. The parse determines whether the PP modifies the seeing event or the noun student.

3.3 Semantic Composition

1. Assign lexical meanings: researcher = person who researches; see = visual perception; student = learner; telescope = optical instrument.
2. Combine “experienced” with “researcher” to form the noun phrase experienced-researcher.
3. Combine “the student” as the object/theme of SEE.
4. Attach “with the telescope” either to SEE (instrument relation) or to student (possession/accompaniment relation).
5. Build the sentence-level event representation.

The interpretations diverge at step 4, when the parser attaches the prepositional phrase to a different constituent.

3.4 Failure Analysis

Error 1: The parser attaches the telescope to student even though the intended meaning is that the researcher used it. Validation: check whether the PP attachment is compatible with the verb’s instrument role and the surrounding context.

Error 2: A system assigns roles only from word order and treats the first noun as the agent in every construction. This can fail with passive voice or unusual word order. Validation: use dependency relations and semantic-role constraints, not position alone.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (9 Marks)

4.1 Lexemes and Senses

A lexeme is an abstract vocabulary item that groups related word forms and senses. A word form is the actual form appearing in a sentence. A sense is one particular meaning of that lexeme.

Context	Word form / lexeme	Part of speech	Selected sense
Nila deposited cash at the bank.	bank / BANK	Noun	Financial institution.
They sat on the bank of the river.	bank / BANK	Noun	Land beside a river.
The pilot banked the aircraft to the left.	banked / BANK	Verb, past tense	Tilt or turn an aircraft.

4.2 Design a WSD Method

Input: a sentence containing bank/banked. Candidate senses: FINANCIAL_INSTITUTION, RIVER_EDGE and AIRCRAFT_TURN. Context window: the target word plus three words on either side and the full sentence when available. Features: nearby nouns and verbs, part of speech, and collocations.

Decision rule: score each candidate sense by matching context words to a small sense dictionary. “cash, deposit, account, money” support FINANCIAL_INSTITUTION; “river, water, shore, sat” support RIVER_EDGE; “pilot, aircraft, plane, left, turn” support AIRCRAFT_TURN. Select the highest score; if scores tie or are weak, return multiple candidates for clarification.

4.3 Test the Method

Sentence	Evidence	Output
Nila deposited cash at the bank.	deposited, cash	FINANCIAL_INSTITUTION
They sat on the bank of the river.	river, sat	RIVER_EDGE
The pilot banked the aircraft to the left.	pilot, aircraft, left	AIRCRAFT_TURN
The bank approved my home loan.	approved, loan	FINANCIAL_INSTITUTION

Possible failure: “The bank was crowded” has little context. It may refer to a financial institution or a river bank in a story. The method should preserve uncertainty rather than force a decision.

4.4 Evaluation

Create a labelled test set of 30 sentences: 10 financial, 10 river-edge and 10 aircraft-turn examples. Two annotators assign the intended sense; disagreements are reviewed and resolved using context. Compare system predictions with the agreed labels.

Accuracy = correct predictions / total predictions × 100. Example: if 27 of 30 are correct, accuracy = (27/30) × 100 = 90%. A balanced test set gives each sense equal representation, but real text may contain mostly one sense. Accuracy may then hide poor performance on rare senses. Report per-sense precision/recall or a confusion matrix as well.

Task 5 – Information Retrieval and Semantic Matching (8 Marks)

5.1 Query and Document Representation

Query: “methods for finding the meaning of ambiguous words”. Central concepts: word meaning, ambiguity, context, sense selection and word sense disambiguation (WSD). Intent: retrieve explanations or methods that identify which meaning of a word is intended in a particular context.

Represent the query and each document using normalised terms, concept labels and (optionally) a semantic vector. Example concepts: WSD, lexical sense, contextual evidence, supervised classification, dictionary-based disambiguation.

5.2 Lexical versus Semantic Retrieval

Keyword retrieval looks for exact or near-exact words such as “ambiguous”, “meaning” and “words”. It is simple and fast but may miss documents using different wording.

Example missed document: “Context-sensitive lexical interpretation using neighbouring tokens.” It may not contain the query’s exact words, but it discusses the same task. Semantic retrieval can find it because its concepts are similar to the query.

5.3 Ranking Design

Use three signals: (1) lexical overlap L, (2) semantic similarity S, and (3) concept coverage C. Define Score = 0.30L + 0.50S + 0.20C, where each signal is normalised between 0 and 1. Semantic similarity receives the largest weight because the query asks for meaning-based matching. Sort by descending score; break ties using higher concept coverage, then document ID.

5.4 Evaluation and Error Analysis

Suppose the system retrieves D1, D2 and D3. D1 and D2 are relevant; D3 is a related but non-answering article. There are four relevant documents in the complete collection.

Measure	Calculation	Result
Precision	Relevant retrieved / all retrieved = 2/3	0.667 (66.7%)
Recall	Relevant retrieved / all relevant = 2/4	0.50 (50%)

A semantic-retrieval error is a false positive: an article about general dictionary definitions may be related to word meaning but may not explain how to choose a sense in context. Improve the system by adding a task-specific concept signal for contextual sense selection and checking relevance with a labelled validation set.

Task 6 – Integrated Semantic Analysis Challenge (8 Marks)

Stage 1 – Interpret the Problem

Sentence: “I saw her duck near the bank, and it was closed.” Ambiguities: duck can be a bird (noun) or an action of lowering the head/body (verb); bank can be a financial institution or river edge; “it” may refer to bank or another singular entity. The word “closed” strongly supports a building or institution reading of bank, but context should still be checked.

Stage 2 – Build Candidate Interpretations

Candidate A: The speaker saw her bird near a financial bank building, and the bank was closed. Senses: duck = bird; bank = financial institution; it = bank.

Representation: SEE(e1) ∧ Experiencer(e1,speaker) ∧ Theme(e1,duck_bird) ∧ Near(duck_bird,bank_building) ∧ Closed(bank_building).

Candidate B: The speaker saw her lower her body near a river bank; “it” refers to a separate object or place understood from prior context. Senses: duck = action; bank = river edge. This reading needs additional discourse context because “it was closed” does not naturally describe a river bank.

Representation: SEE(e1) ∧ Experiencer(e1,speaker) ∧ Theme(e1,duck_action) ∧ Near(duck_action,river_bank) ∧ Closed(x), where x must be resolved from earlier context.

Stage 3 – Syntax-Driven Analysis and WSD

The parser checks whether "duck" is a noun phrase object of saw or a verb in a clause such as "saw her duck". The possessive/determiner interpretation and phrase structure determine the reading. Context words "closed" and the pronoun "it" favour a bank institution. The system should retain alternatives or ask "Do you mean a financial bank or a river bank?" if confidence is low.

Stage 4 – Information Retrieval Trace

Selected query concepts: financial institution, opening hours, closure status, bank branch. Example ranking:

Document	Description	Illustrative score	Decision
D1	How to check whether a bank branch is open or closed	0.94	Relevant – top result
D2	Bank branch opening hours and holiday closures	0.89	Relevant
D3	Bird species and duck habitats near rivers	0.35	Wrong sense
D4	River bank erosion and environmental management	0.22	Wrong sense

The query includes the selected sense FINANCIAL_INSTITUTION and the predicate CLOSED, so D1 and D2 rank above river-related documents even though the latter share the word "bank" or "duck".

Stage 5 – Challenge and Refine

Misleading context: "The duck was beside the river bank, and it was closed to visitors." A simple keyword method may select a bank-branch closure article. The system should recognise "river" and "visitors" as evidence for a natural area. Refine it by adding entity-type compatibility and discourse-level pronoun resolution before retrieval.

PART II – SEMANTIC SYSTEMS ENGINEERING: ANALYSE, REPRESENT, TEST, DISAMBIGUATE & RETRIEVE

Part II extends the same semantic reasoning into reproducible traces, finite models, WSD test sets and ranked retrieval. The scores and test data below are illustrative hand-worked examples, not outputs from an actual software run.

Task 1 – Representing Meaning and Language Structure (8 Marks)

1.1 Semantic Structure Analysis

Sentence: "Leena sent the report to Arun after the meeting." Event = SEND. Agent = Leena. Theme = the report. Recipient = Arun. Temporal expression = after the meeting. Implicit information: the sentence does not specify the exact meeting, exact sending time, delivery channel or whether Arun read the report.

Surface form is the English subject–verb–object–prepositional phrase order. The represented meaning records event roles and a temporal relation independently of that order.

1.2 Build Meaning Representations

A. Semantic-role/event frame:

Slot	Value
Event	e1
Predicate	SEND
Agent	Leena
Theme	report
Recipient	Arun
Time relation	AFTER(e1, meeting1)

B. Graph / slot representation:

Nodes: Leena (Person), e1 (SendEvent), report1 (Document), Arun (Person), meeting1 (Meeting).
Edges: Agent(Leena,e1); Theme(e1,report1); Recipient(e1,Arun); After(e1,meeting1).

The frame is compact and easy to query. The graph explicitly shows how the event and entities are connected and can be extended with additional facts.

1.3 Meaning-Preserving Variation

Alternative: "After the meeting, Leena sent Arun the report." The order changes, but Agent=Leena, Theme=report, Recipient=Arun and After(send_event, meeting) remain unchanged. Meaning changes if the recipient and theme are swapped, for example "Leena sent Arun to the report," because the roles are no longer the same.

1.4 Representation Check

Validation rule: every SEND event must have exactly one Agent and one Theme; if a Recipient is expressed, it must be a Person or valid destination entity. If Agent or Theme is missing, flag the representation for review.

Task 2 – First Order Predicate Calculus: Formalise and Test (9 Marks)

2.1 Predicate Vocabulary

Symbol	Meaning
Learner(x)	x is a learner.
Course(c)	c is a course.
Enrolled(x,c)	learner x is enrolled in course c.
Completed(x,c)	learner x completed course c.
Prerequisite(p,c)	course p is a prerequisite for course c.
OfferedBy(c,d)	course c is offered by department d.
NLP	Constant representing the NLP course.
Dept	Constant representing the department.

2.2 Formalisation

(i) Every learner enrolled in NLP has completed at least one prerequisite:

$\forall x\ [(\text{Learner}(x) \wedge \text{Enrolled}(x,\text{NLP})) \rightarrow \exists p\ (\text{Course}(p) \wedge \text{Prerequisite}(p,\text{NLP}) \wedge \text{Completed}(x,p))]$

(ii) Some learner completed every course offered by the department:

$\exists x\ [\text{Learner}(x) \wedge \forall c\ ((\text{Course}(c) \wedge \text{OfferedBy}(c,\text{Dept})) \rightarrow \text{Completed}(x,c))]$

(iii) No learner completed a course that they were not enrolled in:

$\forall x \forall c\ [(\text{Learner}(x) \wedge \text{Course}(c) \wedge \text{Completed}(x,c)) \rightarrow \text{Enrolled}(x,c)]$

2.3 Quantifier-Scope Challenge

For (ii), $\exists x$ has scope over the entire universal statement: there is one particular learner who completed every department course. Compare $\forall c \exists x$ Completed(x,c): for each course, possibly a different learner completed it.

Model	Facts	Truth values
Learners A, B; courses C1, C2	Completed(A,C1), Completed(B,C2) only	$\exists x \forall c$ is false; $\forall c \exists x$ is true.

The formulae are not equivalent. The first requires one learner to complete all courses; the second only requires each course to have at least one learner who completed it.

2.4 Consistency Test

Tiny knowledge base:

- Learner(Anu).
- Course(NLP).
- Course(Math).
- Enrolled(Anu,NLP).
- Completed(Anu,NLP).
- Rule: ∀x∀c [(Learner(x) ∧ Completed(x,c)) → Enrolled(x,c)].

Query: Does Enrolled(Anu,NLP) follow? Yes. It is directly present as a fact. A second query, Enrolled(Anu,Math), does not follow because neither a fact nor the rule derives it. The rule derives enrolment only when completion is known; no completion fact for Math is provided. The knowledge base is consistent under the stated facts.

Task 3 – Linguistic Concepts and Syntax-Driven Semantic Analysis (9 Marks)

3.1 Linguistically Relevant Concepts

Sentence: "The analyst noticed the student with the camera near the library." Entities: analyst, student, camera, library. Predicate: NOTICE. Modifiers: the, the; "near the library" is a location phrase. "With the camera" may attach to student or to the noticing event. Relevant roles: experiencer/agent = analyst; theme = student; instrument = camera in the instrument reading; location = library.

3.2 Parse-to-Meaning Procedure

Parse A – "with the camera" modifies student: [noticed [the student [with the camera]] [near the library]]. Meaning: the analyst noticed a student who had a camera; the noticing occurred near the library.

Representation: NOTICE(e1) ∧ Experiencer(e1,analyst) ∧ Theme(e1,student) ∧ Has(student,camera) ∧ Location(e1,library).

Parse B – "with the camera" modifies noticed: [noticed [the student] [with the camera] [near the library]]. Meaning: the analyst used the camera while noticing the student near the library.

Representation: NOTICE(e1) ∧ Experiencer(e1,analyst) ∧ Theme(e1,student) ∧ Instrument(e1,camera) ∧ Location(e1,library).

6. Identify noun phrases and the main verb noticed.
7. Create the event NOTICE(e1) with analyst as experiencer and student as theme.
8. Attach "with the camera" to either student (Has) or event (Instrument).
9. Attach "near the library" to the event as its location, unless wider context supports another attachment.

The two readings diverge when "with the camera" is attached.

3.3 Composition and Type/Role Checks

Word meanings combine into noun phrases and then into the event. The adjective/noun combination "the analyst" identifies the experiencer; "noticed the student" builds the core event; prepositional phrases add relations.

Check 1: Instrument compatibility – an instrument relation should connect an event to a usable instrument, such as a camera. Check 2: Entity-type and attachment compatibility – Has(student,camera) requires a plausible possession relation, while Location(e1,library) requires a place entity.

3.4 Error Challenge

A syntactically possible but implausible reading is that the library is the instrument used for noticing: Instrument(e1,library). The system may produce this if it attaches every nearby prepositional phrase to the verb and ignores semantic types. Refine the parser with selectional preferences: instruments are typically tools, while libraries are locations; context and entity types should penalise the implausible reading.

Task 4 – Lexemes, Senses and Word Sense Disambiguation (8 Marks)

4.1 Lexeme and Sense Inventory

Sentence	Word form / POS	Sense
Turn on the light.	light – noun	A lamp or source of illumination.
This bag is light.	light – adjective	Having little weight.
The light meal was served early.	light – adjective	Not heavy or substantial; easy to digest.

Lexeme LIGHT groups related word forms; the word form is the occurrence “light”; the sense is the context-specific meaning. The noun and adjective uses have different grammatical functions.

4.2 WSD Design

Input: sentence and target token light. Candidate labels: LIGHT_SOURCE, LOW_WEIGHT and NOT_HEAVY_FOOD. Features: part of speech, nearby words, dependency relation and collocations. Decision: noun with “turn on”, “lamp” or “room” → LIGHT_SOURCE; adjective modifying “bag”, “box” or “object” → LOW_WEIGHT; adjective modifying “meal”, “food” or “snack” → NOT_HEAVY_FOOD. If evidence conflicts, return uncertain.

4.3 Apply and Challenge

Sentence	Evidence	Predicted sense
Turn on the light.	Verb phrase “turn on”; object noun	LIGHT_SOURCE
This bag is light.	Adjective after linking verb; subject bag	LOW_WEIGHT
The light meal was served early.	Adjective modifies meal	NOT_HEAVY_FOOD
She packed a light suitcase for the trip.	suitcase, packed	LOW_WEIGHT

Failure case: “The room has a light colour.” The word light may mean pale in colour, a sense absent from the three-label inventory. The system should flag an unknown/other sense and expand the inventory rather than force a wrong label.

4.4 Evaluation

Illustrative labelled test set (8 examples):

No.	Sentence (short form)	Gold label	Prediction
1	Turn on the light.	LIGHT_SOURCE	LIGHT_SOURCE
2	The light is broken.	LIGHT_SOURCE	LIGHT_SOURCE
3	This bag is light.	LOW_WEIGHT	LOW_WEIGHT
4	A light suitcase is easy to carry.	LOW_WEIGHT	LOW_WEIGHT
5	We ate a light meal.	NOT_HEAVY_FOOD	NOT_HEAVY_FOOD
6	They served light food.	NOT_HEAVY_FOOD	NOT_HEAVY_FOOD
7	The light meal was ready.	NOT_HEAVY_FOOD	LOW_WEIGHT
8	Turn off the light.	LIGHT_SOURCE	LIGHT_SOURCE

Accuracy = 7 correct / 8 total × 100 = 87.5%. Annotation policy: label the sense intended by the full sentence; if two readings remain equally possible, mark ambiguous and exclude from the primary accuracy calculation while reporting the count separately. Class imbalance can make overall accuracy misleading because a model may perform well on the most common sense but poorly on rare senses. Report per-class results as well.

Task 5 – Semantic Information Retrieval (8 Marks)

5.1 Query and Document Semantics

Query: “How can a system identify the intended meaning of a word in context?” Intent: find methods for contextual word-sense disambiguation. Concepts: target word, context, candidate senses, contextual clues, sense classification and WSD.

ID	Candidate document	Relevance
D1	A practical guide to word sense disambiguation using context	Relevant
D2	Contextual lexical interpretation with neighbouring words	Relevant
D3	Supervised classification of dictionary senses	Relevant
D4	A list of common English words and their spellings	Not relevant

D5	How to build a search engine index using term frequency	Not relevant
D6	Polysemy and choosing a word meaning in a sentence	Relevant

5.2 Ranking Strategy

Use lexical overlap L and semantic similarity S, both scaled from 0 to 1. Score(D) = 0.35L + 0.65S. Semantic similarity has the higher weight because documents may explain the same idea using different words. Ties are resolved by higher concept coverage, then document ID.

5.3 Retrieval Trace

Document	**L**	**S**	**Score = 0.35L + 0.65S**	**Rank**
D1	0.90	0.95	0.9325	1
D2	0.35	0.92	0.7205	2
D3	0.65	0.80	0.7475	2 (above D2)
D4	0.30	0.15	0.2025	5
D5	0.55	0.20	0.3225	4
D6	0.40	0.88	0.7120	3

Final order: D1, D3, D2, D6, D5, D4. D2 has low literal overlap but high semantic similarity, so it ranks above keyword-heavy distractors D4 and D5.

5.4 Evaluate

At cutoff k=4, retrieved documents are D1, D3, D2 and D6; all four are relevant. Precision@4 = 4/4 = 1.00 (100%). There are four relevant documents in the collection, so Recall@4 = 4/4 = 1.00 (100%).

To illustrate errors, suppose a broader query adds D5 and misses D6: D5 is a false positive because it discusses search indexing rather than word meaning; D6 is a false negative if the system fails to retrieve it. Improve by adding concept coverage for “intended sense in context” and testing against labelled relevant and distractor documents.

Note: the above ranking values are illustrative. In a real system, lexical and semantic scores would be computed from a defined retrieval model and evaluated on a larger labelled collection.

Task 6 – Integrated Semantic Analysis Design Challenge (8 Marks)

6.1 Model the Ambiguity

Sentence: “I watched the crane by the bank, then it moved.” Ambiguities: crane may be a construction machine or a bird; bank may be a financial institution or the land beside a river; “it” may refer to crane or bank. Useful context includes construction-site vocabulary, bird/nature vocabulary, nearby verbs, location and the discourse referent.

6.2 Candidate Meaning Structures

Candidate A – construction scene: crane = lifting machine; bank = financial institution or building near the site; "it" = crane. Roles: watcher = I; watched entity = crane machine; location = near bank. Representation: WATCH(e1) ∧ Experiencer(e1,I) ∧ Theme(e1,crane_machine) ∧ Near(crane_machine,bank_building) ∧ MOVED(crane_machine).

Candidate B – nature scene: crane = bird; bank = river edge; "it" = bird. Representation: WATCH(e1) ∧ Experiencer(e1,I) ∧ Theme(e1,crane_bird) ∧ Near(crane_bird,river_bank) ∧ MOVED(crane_bird).

6.3 Syntax and WSD Trace

The parser identifies "the crane" as the object of watched and "by the bank" as a location phrase. Syntax alone does not decide the senses. Contextual clues such as "steel", "operator", "lifting" support the machine; "wings", "nest", "river" support the bird and river bank. Pronoun resolution links "it" to the most plausible recent singular entity, usually crane. If evidence is insufficient, retain both readings or ask whether the user means a machine or a bird.

6.4 Semantic Retrieval

For Candidate B, query concepts are crane bird, river bank, bird movement and habitat. For Candidate A, query concepts are construction crane, lifting equipment, machine movement and site safety. Example ranking for a nature-scene selection:

Document	Description	Illustrative score	Result
D1	Crane birds: habitat and movement near wetlands	0.96	Relevant
D2	Bird identification and behaviour of cranes	0.90	Relevant
D3	Construction crane operating and safety guide	0.28	Wrong sense
D4	River bank restoration and wildlife habitat	0.78	Related context

Ranking: D1, D2, D4, D3. D3 contains the exact word "crane" but refers to machinery, so sense compatibility lowers its score. D4 is related to river habitat but may not directly explain crane birds.

6.5 Adversarial Refinement

Misleading input: "I watched the crane by the bank while the operator lifted steel, then it moved." A system relying only on the word "bank" may choose a river setting. The intended goal is not met if it selects bird documents. Refine the model by using multiple contextual features ("operator", "lifted", "steel"), entity-type constraints and coreference scoring. Do not let one ambiguous keyword dominate the decision.

Conclusion

Semantic NLP systems should preserve the meaning of a sentence through explicit events, semantic roles, logical formulae and structured relations. Ambiguity must be represented through alternative parses or senses and resolved using syntax, context and entity compatibility. Information retrieval should combine lexical and semantic evidence, then be evaluated with measurable metrics. A reliable system should retain uncertainty, test difficult cases and refine errors instead of accepting every generated answer as correct.

Quick Glossary

Term	Simple meaning
Semantic role	The part played by an entity in an event, such as agent, theme or recipient.
FOPC	First-order predicate calculus; a formal language using predicates, variables and quantifiers.
Lexeme	An abstract vocabulary item that can have different word forms and senses.
Word sense	A particular meaning of a word in a context.
WSD	Word Sense Disambiguation; selecting the intended meaning of a word.
Semantic retrieval	Finding documents based on meaning, not only exact keyword overlap.
Precision	How many retrieved documents are relevant.
Recall	How many of all relevant documents were retrieved.
Coreference	Linking expressions such as “it” to the entity they refer to.